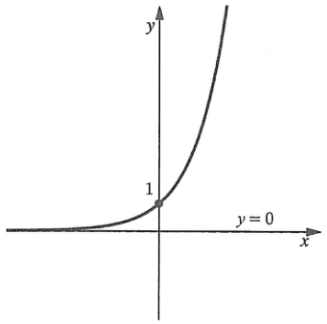
Graph Sketching: e and Natural Logarithms

Questions 1–26

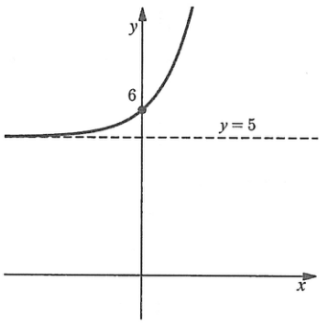
Question 1. Find the equation of each of the following graphs in the form

$$y = e^x + b$$

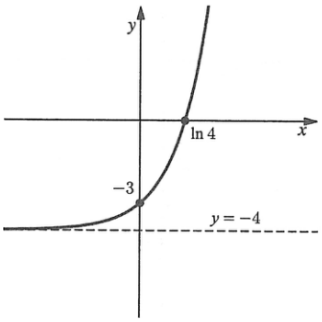
a)



b)



c)



Question 2. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

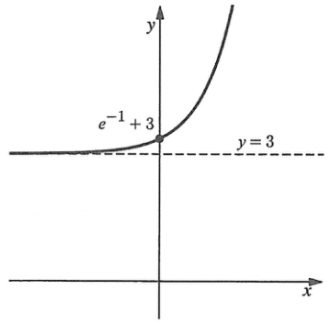
a) $y = e^x - 6$

b) $y = e^x + e$

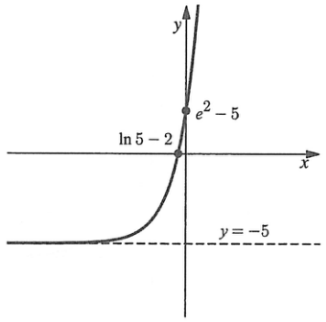
Question 3. Find the equation of each of the following graphs in the form

$$y = e^{x-a} + b$$

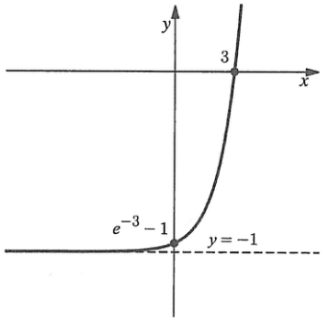
a)



b)



c)



Question 4. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

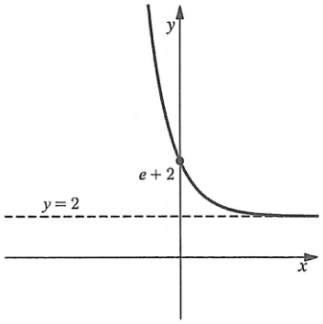
a) $y = e^{x+1} - 10$

b) $y = e^{x-e} - e$

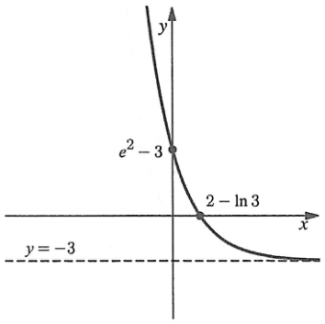
Question 5. Find the equation of each of the following graphs in the form

$$y = e^{a-x} + b$$

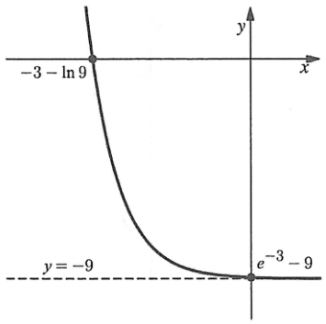
a)



b)



c)



Question 6. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

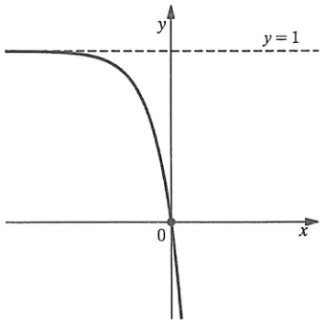
a) $y = e^{5-x} - 20$

b) $y = e^{-1-x} + 20$

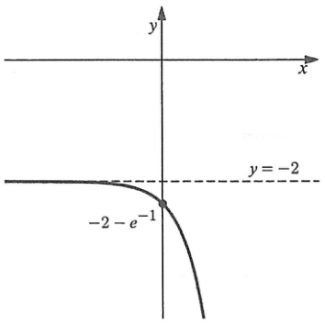
Question 7. Find the equation of each of the following graphs in the form

$$y = b - e^{x-a}$$

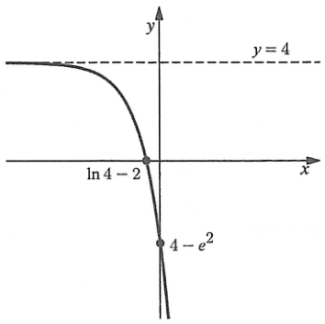
a)



b)



c)



Question 8. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

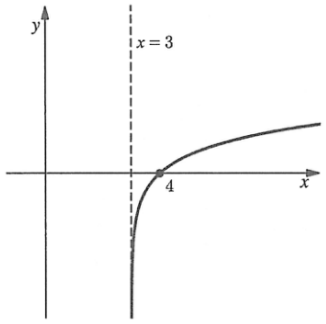
a) $y = 10 - e^{x+4}$

b) $y = 5 - e^{4-x}$

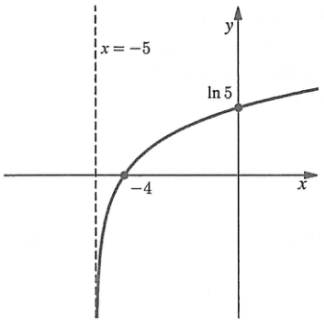
Question 9. Find the equation of each of the following graphs in the form

$$y = \ln(x - a)$$

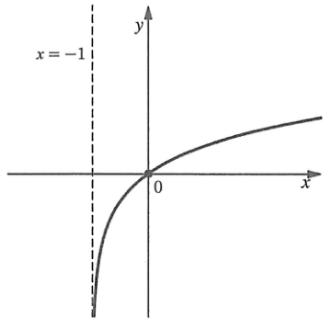
a)



b)



c)



Question 10. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

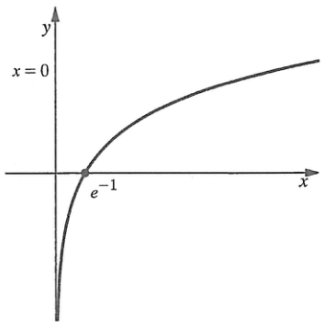
a) $y = \ln(x - 2)$

b) $y = \ln(x + 2)$

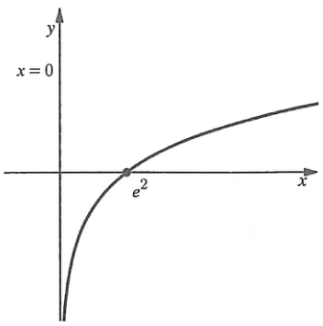
Question 11. Find the equation of each of the following graphs in the form

$$y = \ln x + b$$

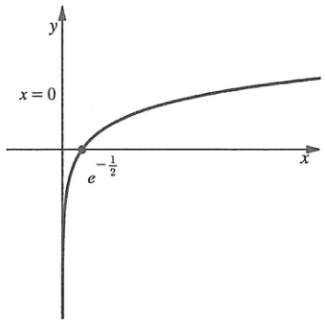
a)



b)



c)



Question 12. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

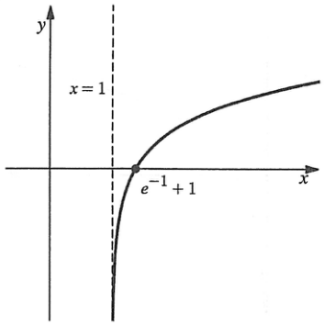
a) $y = \ln x + 3$

b) $y = \ln x - e$

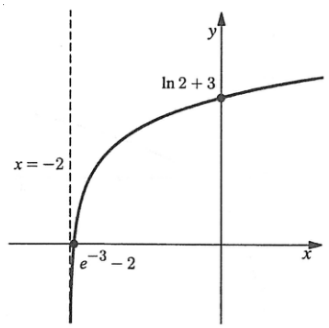
Question 13. Find the equation of each of the following graphs in the form

$$y = \ln(x - a) + b$$

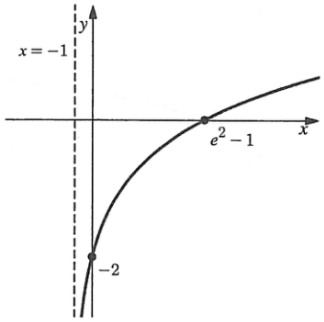
a)



b)



c)



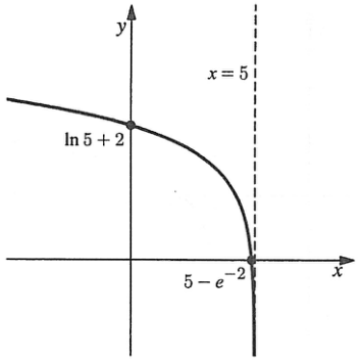
Question 14. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \ln(x - 3) - 3$

b) $y = \ln(x - e) + e$

Question 15. Find the equation of the following graph in the form

$$y = \ln(a - x) + b$$

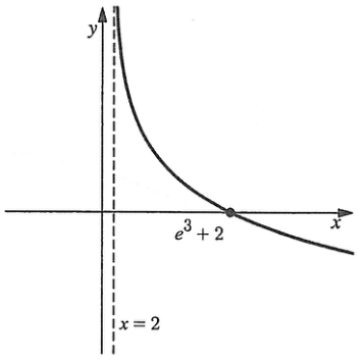


Question 16. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \ln(8 - x) - 1$
b) $y = \ln(1 - x) + 3$
c) $y = \ln(e - x) - e$
d) $y = \ln(-e - x) + 2e$

Question 17. Find the equation of the following graph in the form

$$y = b - \ln(x - a)$$



Question 18. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = 1 - \ln(x - 1)$
b) $y = e - \ln(x + e)$
c) $y = 1 - \ln(2 - x)$
d) $y = \frac{e}{2} - \ln(3e - x)$

Question 19. Prerequisite knowledge for Q19-Q22: Exponentials and Logarithms Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = 2^x$ and $y = e^x$
b) $y = 3^x$ and $y = e^x$
c) $y = e^x$ and $y = e^{-x}$

Question 20. Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = -2^x$ and $y = -e^x$
b) $y = 3^x$ and $y = -e^x$
c) $y = -e^{-x}$ and $y = -e^x$

Question 21. Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = \log_2 x$ and $y = \ln x$

b) $y = \log_3 x$ and $y = \ln x$

c) $y = \log_2(-x)$ and $y = \ln(-x)$

Question 22. Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = -\log_3 x$ and $y = -\ln x$

b) $y = -\log_2 x$ and $y = \ln x$

c) $y = -\log_3(-x)$ and $y = -\ln(-x)$

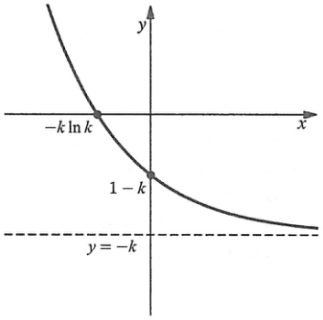
Question 23. Given that $k > 1$, sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = e^{kx} - 2k$

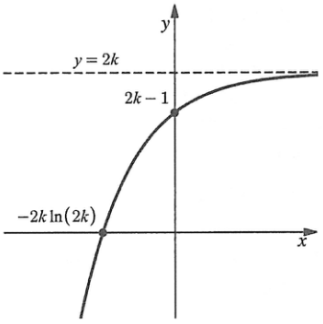
b) $y = k - e^{-kx}$

Question 24. Given that $k > 1$. Write down a suitable equation for each of the following curves.

a)



b)



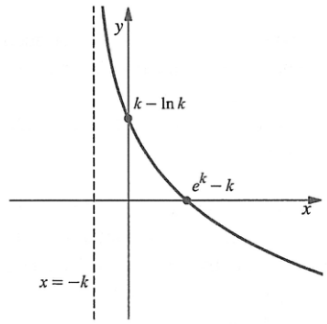
Question 25. Given that $k > 1$, sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = k - \ln x$

b) $y = k\ln(x + k) - k$

Question 26. Given that $k > 1$, write down a suitable equation for each of the following curves.

a)



b)

